

Two Origins

Continuation, the Fencepost Tax, and Where Computational Difficulty Is Located

Abstract

A continuation principle can be written two ways. It can say that the successor of an admissible state is itself admissible — that transformation never leaves the space in which transformation is possible — or it can say that no state is permitted to be its own only continuation. The two readings are usually treated as the same sentence. They are not. The first prunes nothing: measured across all total maps on a finite state space, it excludes exactly zero candidate laws, because it is a typing statement and a type declaration should be free. The second is the entire content of the principle, and its price is exactly $\log_2(e) = 1.442695$ bits, asymptotically independent of the size of the space.

That price has a source. The maps invariant under a change of index origin are precisely the translations, and exactly one of them is excluded by the second reading: the translation by zero, which is the unique map under which the two origins coincide. The forbidden object is not a state that fails to move. It is the collapse of two simultaneous readings of one object into one reading. The per-state cost of keeping them apart is $\log_2(n/(n-1))$, which is identically the continuation tax $\log_2((k+1)/k)$ at $k = n-1$, and n copies of it converge to $\log_2(e)$. Fencepost, tax, and wall are the same quantity.

The remainder of the paper is a sequence of exhaustive or fully seeded measurements in which a task is posed, a cost is paid, and the location of that cost is audited. In every case the cost is found outside the object. Observation of a hidden transition law saturates after eight steps and never improves again, with a residual floor that decomposes exactly into an off-path term matching a closed form to all reported digits and an on-path term produced by the projection. The information needed to locate a target is conserved at $\log_2(n)$ and moves freely between the arrangement and the reader without ever residing in the target. A sign gradient carries direction and exactly zero bits about distance at every scale tested, and a bisection over 4096 positions changes nothing in the field while appending precisely $\log_2(n)$ entries to the reader's history.

Three results are surprises against the author's own prior statements. Hardening a search landscape does not remove the easy directions: after a single round of nonlinear mixing roughly forty percent of local moves still strictly improve, while the success rate falls from forty out of forty to zero. What is destroyed is composability, not availability, and it is destroyed in one round rather than along a ramp. A fixed-capacity accumulator changes $k/2$ bits of state on every added quantum regardless of how full it is, so capacity is exercised completely from the first addition and there is no fill gradient. And the three-dimensional checkerboard — the maximum of local disorder at 1.0000 — compresses to 160 bytes and separates under a three-bit rule, while a random field sits at local disorder 0.5031 and compresses to

5322 bytes. Foldability is high at both extremes of disorder and collapses in the middle, which means mixedness and description length are maximized in different places and are not the same quantity.

Two closing measurements bound what work can reach. Random pairwise stirring saturates at local disorder 0.50 and remains statistically indistinguishable from two hundred thousand strokes through four million; the perfect mix at 1.00 is not merely slow to reach but unreachable by stirring, and is available only as a rule. Exhaustive breadth-first search of the $2 \times 2 \times 2$ permutation group finds 3,674,160 states, a diameter of eleven, and 51.4 percent of the entire space at distance exactly nine. Disorder does not purchase distance. The paper claims a relocation of the question of computational difficulty, not its resolution, and states explicitly where the relocation does not reach.

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1. Two accounts of continuation

Let X be a set of states and T a transformation on it. A continuation principle asserts that a state cannot simply stop. There are two natural ways to write that assertion and they are routinely treated as paraphrases of one another.

The first is closure. It says that the codomain of an admissible transformation is its domain, so that nothing an admissible transformation produces can fall outside the space where transformation is defined:

$$T : X \rightarrow X \quad \text{and for a family, for all } T \text{ in } \Gamma, \quad T(X) \subset X$$

Read this way, continuation is not a property that a produced state must acquire. The state's existence as an output of an admissible transformation is already its admissibility. There is no validation event, no checker sitting outside the system asking whether the new state qualifies. This is an attractive formulation and it removes a real conceptual defect: it stops the theory from smuggling in an external adjudicator.

The second is what we will call the wall condition. It says that no state may be its own only continuation:

$$x \in X \Rightarrow \text{there exists } T \text{ in } \Gamma \text{ with } T(x) \neq x$$

These look like the same idea with different emphasis. The purpose of the next two sections is to show that they are not, that only one of them does any work, and that the one which does work turns out to be a statement about index conventions rather than about motion.

2. Closure prunes nothing

A constraint on a theory is worth what it excludes. If a candidate law is admissible under a constraint no matter what the law does, the constraint has told us nothing about laws. This is measurable directly: enumerate every total map on a finite X and count how many satisfy closure.

n	total maps	satisfy closure	fraction	bits pruned
1	1	1	1.0000	0.0000
2	4	4	1.0000	0.0000
3	27	27	1.0000	0.0000
4	256	256	1.0000	0.0000
5	3125	3125	1.0000	0.0000
6	46656	46656	1.0000	0.0000
7	823543	823543	1.0000	0.0000

Table 1. Closure as a filter on candidate laws. Every total map on X lands in X by the definition of a total map. The exclusion is empty at every n . [THEOREM, trivial]

Zero bits, at every size. This is not a defect in the measurement. It is the correct reading of what closure is: a typing statement. It fixes what kind of object a law is — an endomorphism of the state space rather

than a partial map into some larger ambient set — and a type declaration should cost nothing, because it is not making a claim about which laws are permitted. It is saying there is no outside.

The consequence is that closure cannot be the whole of a continuation principle. Consider the identity map. It is total. It is closed. Every state it produces is admissible, and every produced state is immediately available as the starting condition for the next application. It satisfies the sentence "every generated state is already ready for its next transformation" without qualification. It is also the dead state: nothing distinguishes any successor from its predecessor, in value or in relation.

Closure does not merely fail to exclude the frozen state. Closure is precisely the formulation under which the frozen state is fully admissible.

So the two accounts in Section 1 are not paraphrases. Closure is the type. The wall condition is the law. Both are needed and neither substitutes for the other, and conflating them readmits the exact object the principle exists to forbid.

3. The wall condition and its price

Now enumerate under the second reading. On a finite X , a state whose only continuation is itself is a fixed point of the map. Excluding fixed points everywhere gives the admissible set.

n	total maps	admissible	fraction	bits
1	1	0	0.000000	undefined
2	4	1	0.250000	2.000000
3	27	8	0.296296	1.754888
4	256	81	0.316406	1.660150
5	3125	1024	0.327680	1.609640
6	46656	15625	0.334898	1.578206
7	823543	279936	0.339917	1.556747
8	16777216	5764801	0.343609	1.541161
9	387420489	134217728	0.346439	1.529325

Table 2. The wall condition as a filter. Brute-forced for $n \leq 7$ and checked against $(n-1)^n$ throughout. The fraction converges to $1/e$ and the cost to $\log_2(e)$. [THEOREM]

The admissible fraction is $(n-1)^n / n^n$, which tends to $1/e$, so the cost tends to $\log_2(e) = 1.442695040889$ bits. The number is asymptotically independent of the size of the state space. Whatever this constraint is charging for, it charges the same total no matter how large the world is.

The $n = 1$ row deserves separate attention. There are zero admissible maps on a one-state space, so the wall condition is unsatisfiable there. It follows that the continuation principle, in the only form that does any work, forces the state space to contain at least two states.

C1 in its load-bearing form is unsatisfiable on a one-state space. Continuation therefore forces distinction; C0 is not an independent primitive but the condition under which C1 can be satisfied at all.

This is consistent with an earlier structural result in this programme, namely that writing a constraint cascade as an ordered list is a linearization performed by the reader rather than a feature of the constraint system, which is a simultaneous mutually-forcing fixed point. The ordering C0 then C1 is a presentation choice. The forcing runs in both directions.

The cost decomposes exactly, and not into a new quantity:

n	wall cost (bits)	$n \cdot (-\log_2(1 - 1/n))$	equal
2	2.000000000000	2.000000000000	True
3	1.754887502163	1.754887502163	True
4	1.660149997115	1.660149997115	True
8	1.541160623539	1.541160623539	True
16	1.489750470264	1.489750470264	True
64	1.454084896005	1.454084896005	True
1024	1.443399940783	1.443399940783	True
1000000	1.442695762278	1.442695762278	True

Table 3. The wall cost is n independent copies of the continuation tax $c = -\log_2(1 - f)$ evaluated at occupancy $f = 1/n$. Agreement to $1e-9$ across six orders of magnitude. [MEASURED, exact]

The quantity $-\log_2(1 - f)$ has appeared previously in this programme as the phase-space contraction corresponding to a continuation-width tax of $f = 1/(k+1)$ per symbol, and independently as the mean of $-\log_2|\det J|$ over an invariant measure. It arrives here from a completely different direction: counting maps that refuse to be their own successor. One refusal per state, at occupancy one over the state count, n times.

4. Two origins

The previous section leaves an obvious question. Why should refusing a self-map cost anything at all, and why should the amount be scale-free? The answer is visible if one asks a different question: which laws mean the same thing under a change of index origin?

Any finite ordered structure admits two simultaneous readings. One reads a position as a displacement from an origin and begins at zero. The other reads a position as a count that includes the item itself and begins at one. Both are correct, both describe the same object, and the difference between them is exactly one. Every loop written in every programming language holds both at once without comment: the index runs from zero to $n - 1$ while the count runs from one to n , and the program depends on their coexistence.

Formally, on the cyclic space, changing origin is the shift sigma. A law means the same thing under both conventions exactly when it commutes with sigma.

n	origin-blind maps	are all translations	wall-admissible	= n – 1
2	2	True	1	True
3	3	True	2	True
4	4	True	3	True
5	5	True	4	True
6	6	True	5	True
7	7	True	6	True

Table 4. Maps invariant under change of index origin are exactly the n translations. Exactly one — translation by zero — fails the wall condition. [THEOREM, brute-forced]

The single excluded map is $c = 0$. That is the unique case in which the zero-based reading and the one-based reading of the same object coincide, because displacement by zero is the map under which position and count stop differing.

The wall condition restated: the two origins must not be the same origin. The forbidden object is not a state that fails to move; it is the collapse of two simultaneous readings into one.

This also explains why the $n = 1$ case is empty rather than merely restrictive. On a one-state space there is nowhere for a second origin to differ from the first, so there is no second reading to keep apart, and the constraint has no satisfying assignment. Distinction is not an axiom placed beside continuation. It is the statement that a second origin exists.

The price now reads as what it is. Per state, the wall costs $\log_2(n/(n-1))$, which is the ratio of the count to the index. This is identically the continuation tax at $k = n - 1$:

n	$\log_2(n/(n-1))$	$\log_2((k+1)/k), k = n-1$	equal
2	1.00000000000000	1.00000000000000	True
3	0.584962500721	0.584962500721	True
4	0.415037499279	0.415037499279	True
8	0.192645077942	0.192645077942	True
64	0.022720076500	0.022720076500	True
1024	0.001409570255	0.001409570255	True

Table 5. The per-state wall tax and the continuation tax are the same expression. Agreement to machine precision. [MEASURED]

Three quantities that had been recorded separately are one quantity read three ways: the fencepost difference between counting posts and counting intervals, the continuation tax charged per admissible step, and the exclusion cost of the frozen state. The recurring appearance of a plus-one at the boundary of a completed structure — a wall at support plus one, a carry that does not fit inside the register, a

push and a pull that are one bijection read from opposite ends — is in each case the same second reading demanding its own origin.

5. What continuation buys on a finite space

It is tempting to read a no-terminal-state principle as licensing open-ended novelty. On a finite state space it does not, and the measurement is immediate. A map with no fixed point still has out-degree one everywhere, so every trajectory must eventually revisit a state.

n	admissible maps	have a cycle	contain a 1-cycle
2	1	1	0
3	8	8	0
4	81	81	0
5	1024	1024	0
6	15625	15625	0
7	279936	279936	0

Table 6. Every admissible map on a finite space is eventually periodic, and no admissible cycle has length one. [THEOREM]

On a finite state space, continuation does not purchase novelty. It purchases recurrence of period at least two, necessarily and in every case.

This arrives at a conclusion already reached in this programme from an unrelated direction, where an apparent holographic property of a recurrence was retracted and replaced by recoverability-under-recurrence at a measured recurrence order. Here the same object appears not as an empirical property of a particular recurrence but as a forced consequence of the continuation principle on any finite space.

6. Uniform branching is priced separately

A continuation principle is sometimes taken to imply that every state offers the same number of continuations. It does not. Enumerate the relations on X with out-degree at least one everywhere and no self-loop, and count how many have uniform out-degree.

n	admissible relations	uniform out-degree	fraction	bits
2	1	1	1.000000	0.0000
3	27	9	0.333333	1.5850
4	2401	163	0.067888	3.8807
5	759375	9825	0.012938	6.2722

Table 7. Uniform branching under the continuation and wall conditions. The cost is strictly positive and grows with n. [MEASURED]

Uniform branching is therefore an independent imposition and must be derived or assumed on its own terms. It is not delivered free by continuity, by the wall condition, or — as an earlier experiment in this

programme established with a degree-conserving coupling — by collision. Collision closes the connectivity half of the pair; it demonstrably does not touch the branching distribution, because a coupling that redirects existing edges alters no out-degree by construction.

7. The observation floor

If the object of falsification is the proposed law rather than the produced state, then the natural question is how much a sequence of observations is worth as evidence about a hidden transition relation. Set X to six states, take Γ to be all closed wall-admissible maps, fix a start state, and observe through a three-valued projection: $a(n) = \pi(T^n(x_0))$.

steps observed	distinct observation classes	mean ambiguity (bits)
0	1	13.931569
2	9	10.887712
4	63	8.558865
6	223	7.829282
8	283	7.778238
9	283	7.778238
10	283	7.778238
12	283	7.778238

Table 8. Exhaustive over all 15625 admissible laws. Observation saturates at eight steps and never improves.
[MEASURED]

Six and a fifteen-hundredths bits are purchased, and then nothing further is purchased ever, no matter how long the trajectory runs. The floor is not numerical noise. It decomposes exactly:

component	bits	source
total residual	7.778238	measured
off-path	5.780672	measured
off-path	5.780672	closed form, match True
on-path	1.997565	total minus off-path

Table 9. The floor splits into states the trajectory never entered, whose freedom is $(\text{number unvisited}) \times \log_2(n - 1)$ and matches to all reported digits, and the blur imposed by the many-to-one projection on states that were visited.
[MEASURED]

Every observed value is evidence about the traversed part of the relation, up to the fibres of the projection, and about nothing else. The inverse problem is not underdetermined by insufficient data; it is underdetermined by an exactly computable amount that additional data cannot reduce.

The off-path term is the same $\log_2(n - 1)$ that counted the origin-blind admissible laws in Section 4. The residual freedom of a state the reader never visited is precisely the number of continuations available to it once its own position is excluded.

This is also the quantitative form of a result obtained earlier in this programme by a different method: a single trajectory cannot certify a law off its own orbit, and the residual is a centralizer that no experiment confined to one orbit can close. Collision is the operation that converts off-path freedom into on-path freedom. Here there is a number for how much it has to convert.

8. Locate-cost is conserved

Consider a shelf of n slots, one of which holds the object being sought. The reader must end up knowing which. The question is where the information required to answer that comes from, and the answer is that it can come from either side without changing the total.

n	arrangement	I(appearance ; location)	reader must pay	sum
8	one odd, rest uniform	3.0000	0.0000	3.0000
8	featureless	0.0000	3.0000	3.0000
8	two visible groups	1.0000	2.0000	3.0000
8	four visible groups	2.0000	1.0000	3.0000
64	one odd, rest uniform	6.0000	0.0000	6.0000
64	featureless	0.0000	6.0000	6.0000
1024	one odd, rest uniform	10.0000	0.0000	10.0000
1024	featureless	0.0000	10.0000	10.0000

Table 10. The information required to locate a target is conserved at $\log_2(n)$ and its location is free. [MEASURED]

The locate-cost is never a property of the target. It resides in the arrangement or in the reader, and the sum is fixed.

The familiar experience of finding something instantly because one item on a shelf is turned the wrong way is not a lucky shortcut past a cost. It is the arrangement having already paid it. The reader pays nothing because the shelf spent $\log_2(n)$ on being addressable. A featureless shelf has spent nothing, and the reader owes the whole amount.

9. The gradient carries no distance

A directional signal is the most common form in which an arrangement offers partial help. It is worth measuring what such a signal actually contains.

$n = 16 \quad I(\text{sign}(t - p) ; |t - p|) = 0.000000000000 \text{ bits}$
 $n = 256 \quad I(\text{sign}(t - p) ; |t - p|) = 0.000000000000 \text{ bits}$
 $n = 4096 \quad I(\text{sign}(t - p) ; |t - p|) = 0.000000000000 \text{ bits}$

Mutual information between a sign gradient and distance to the target, at three scales. [MEASURED]

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A sign gradient carries direction and exactly zero bits about how far away the target is, at every scale. This is not a limitation of the measurement; the signal is constant across all distances on one side. Being far from the target is not a fact the field holds. Distance in such a field is not a quantity the field can report, which is a considerably stronger statement than saying it is hard to measure.

If the gradient never reports distance, a reading resolves the target's identity only where the sign is about to change. That window has a fixed width, independent of the size of the space: a bracket of width one or two pins the target, and every wider bracket buys direction only. This is the two-point tangency condition — a single constraint aliases and a pair collides — arriving as a measured property of a search rather than as a design rule for one.

The remaining question is where the cost of a search goes if the field is not changed by it. Bisection over 4096 positions, audited:

```
field entries changed during search : 0
target moved                        : 0
arrangement information gain       : 0.000000
reader history entries appended    : 12
log2(n)                            : 12.000000
```

Where the cost of a search is paid. [MEASURED]

The cost was spent in full, and it is entirely in the length of the reader's history. Nothing else in the system moved.

Twelve entries, and $\log_2(4096) = 12$ exactly. Each entry is one collapse of a two-wide bracket, and the price of one such collapse is $\log_2(2/1) = 1$ bit — the wall tax of Section 3 evaluated at $n = 2$. A search over four thousand positions is twelve repetitions of a single two-unit tangency, and nothing else occurs. Difficulty measured this way is a count of tangencies rather than a magnitude attached to the object.

10. A lower bound that belongs to the reader

It is standard to fix a machine model, derive a bound within it, and report the bound as a property of the problem. The comparison-sorting bound is the clearest case, and it is worth stating the arithmetic plainly because it is not controversial.

n	comparisons	n log2 n	digit reads
1024	8975	10240	4096
8192	96163	106496	32768
65536	965681	1048576	262144
262144	4386881	4718592	1048576

Table 11. Identical sorted output by two methods. Comparisons track $n \log n$; reading the objects' own digits is linear in n . [MEASURED]

Both columns are correct. The $n \log n$ bound is a true theorem about readers that agreed in advance to obtain all their information through pairwise comparison, and it is false as a statement about sorting.

Radix does not defeat the bound; it declines the premise, by reading structure that was present in the objects the whole time.

A lower bound derived inside a fixed model is a statement about the model. Attributing it to the object files reader-side cost as a field property.

The honest boundary must be stated here rather than left implied. This critique lands on comparison bounds, decision-tree arguments and relativized constructions. It does not land on the P versus NP statement itself, which quantifies over all polynomial-time machines and therefore admits any reader that can be built within budget. If the critique applied there, the problem would have closed decades ago. What does survive at that level is narrower and is the subject of the next three sections: the accounting has a column for the reader's work and no column for the arrangement's.

11. Basis, not information

Take a landscape in which local improvement is available and the optimum is reachable by hill-climbing, then apply an invertible mixing to the coordinates and repeat with the identical reader.

basis	solved out of 50	mean local reads
separated (identity)	50	69.6
mixed (random GF(2))	0	26.2

Table 12. Sixteen bits, 65536 states, same optimum, same reader. Only the basis differs. [MEASURED]

The mixing is a bijection. No bit was destroyed and the optimum did not move. What changed is that the information which had been carried by the arrangement — the property that improving one coordinate does not damage the others — is now owed by the reader. This is the same column exchange as Section 8, performed here by an invertible map rather than by removing features.

This is the shape of a musical instrument's difficulty and of a hardened material's. A recorder presents approximately one gradient per degree of freedom; a guitar superposes them so that the local move improving one dimension degrades several others. Neither instrument contains more information than the other. The difference is which basis the player's available moves are expressed in.

It is also the shape of a cryptographic mixing function, and it locates the difficulty differently than is usual. In an earlier result of this programme the rounds of a standard hash were shown to be one-to-one and the message schedule likewise, with the single genuine forgetting event isolated to the feed-forward. The present measurement says that the difficulty of inversion was never located at the forgetting event. It is a basis property of an information-preserving mixing.

12. The ring lock

If the difficulty is a basis, then the key is the de-mixing transformation. The key is not a negative image of the lock. It is the set of moves each of which discharges exactly one ring, verified directly: each

column of the inverse mixing flips exactly one output coordinate. A mould of the lock would be the mixing itself; the key is its inverse, which is a different object.

The interesting quantity is what a partial key buys.

rings aligned	solved out of 40	mean local reads
0	0	25.1
4	0	28.1
8	3	41.8
12	2	56.9
14	9	65.4
15	26	68.2
16	40	67.8

Table 13. Sixteen rings. Half the key returns 7.5 percent; fifteen of sixteen returns 65 percent. [MEASURED]

The value of a key is not additive in its parts. Half of it buys almost nothing, and the transition is concentrated in the final ring.

A lock with one pin picked is locked. This is the mechanical statement of a property that has appeared repeatedly in this programme as an all-at-once condition: a constraint cascade is a simultaneous mutually-forcing system rather than a sequence, and partial satisfaction of a simultaneous system is not partial success.

An honest limitation applies to this particular lock. The mixing used is linear over $GF(2)$, so its inverse is recoverable by elimination in cubic time. The measurement establishes the shape of the threshold, not the existence of hardness; a lock that opens itself is still a lock for the purpose of measuring how partial alignment pays, but it is not evidence that any lock is hard.

13. Hardening preserves the easy directions

The natural hypothesis about a hardened landscape is that hardening removes the local moves that help. It is testable and it is false.

mixing rounds	informative local moves	solved out of 40
0	50.00%	40
1	36.17%	1
2	40.27%	0
3	39.89%	0
4	40.56%	0
5	39.00%	0
6	41.83%	0

mixing rounds	informative local moves	solved out of 40
7	45.28%	0
8	41.64%	0

Table 14. A local move is informative if it strictly reduces distance to the target. Availability is essentially unchanged; success collapses. [MEASURED]

Hardening does not remove local improvement. Roughly forty percent of single-step moves still strictly improve at every depth tested, and the success rate is zero. What is destroyed is the composability of local improvements, not their availability.

This is the correct account of work-hardening in a metal, and not by analogy. Hardened steel retains its slip systems; the dislocations are still present and still slip. What is removed is the ability of a slip to propagate through the piece, because each is pinned by the others. Annealed and hardened material have the same local moves available and only one of them permits a move to run.

The transition is also a quench rather than a ramp: forty out of forty to one out of forty in a single round, then flat. There is no gradient of hardness to descend. Together with Section 12 this gives a consistent statement — partial alignment buys nothing and partial progress buys nothing — and identifies the resource in both cases as composability, which is added or removed whole.

14. The bucket

A container of fixed capacity accumulating discrete quanta is the ordinary picture of a gradient being filled. The measurement contradicts the picture.

quanta accumulated	state bits changed by the next quantum
1	8.030
2	8.000
4	8.110
8	7.977
12	8.020
16	7.757

Table 15. A sixteen-bit accumulator under a nonlinear update. The response is flat at $k/2 = 8.0$ for every fill level. [MEASURED]

The last quantum changes exactly as much state as the first. Capacity is exercised completely from the first addition; there is no fill gradient and no taper.

The apparently smooth filling that motivates the gradient picture is a property of the level — a one-number projection — and not of the state. Watching the level yields a ramp. Watching the state yields half the register every time, forever. The gradient is entirely in the aperture of the reading, which is the same diagnosis reached in Section 9 for distance and in Section 8 for locatability.

15. Three quantities called disorder

A three-dimensional lattice of two colours admits at least three distinct measurements that ordinary language calls the same thing: how locally mixed it is, how long a description it requires, and how much it costs to separate. They are maximized in different places.

field	local disorder	compressed bytes	separation key
uniform, unmixed	0.0000	52	0 bits
perfect mix (checkerboard)	1.0000	160	3-bit rule
random, well stirred	0.5031	5322	32768 bits

Table 16. A 32-cube, 32768 cells. Local disorder is the fraction of adjacent pairs that differ, over all three axes. [MEASURED]

The checkerboard achieves local disorder 1.0000 — every neighbour differs, in every direction, which is the maximum the lattice permits. A well-stirred random field achieves 0.5031. Random paint is half as mixed as perfect mix. At the same time the checkerboard compresses to 160 bytes and separates completely under a single exclusive-or against a three-bit rule, while the random field compresses to 5322 and requires a mask as large as itself.

Foldability is high at both extremes of the disorder axis and collapses in the middle. Order and perfect disorder lie on the same side; randomness is not an extreme of anything but the interior.

Perfect mixedness and zero separation cost are therefore the same state, which resolves an intuition that ordinarily sounds paradoxical. The descent away from that state is steep:

fraction of cells flipped	local disorder	compressed bytes
0.00	1.0000	160
0.02	0.9574	1510
0.10	0.8160	3582
0.25	0.6248	4956
0.50	0.5025	5339

Table 17. Two percent deviation from the fold costs roughly a tenfold increase in description length. [MEASURED]

A random binary string is therefore not a perfect mix. It is what remains after a perfect mix has been fallen away from, and the residue that will not fold is the program. Hardness in the sense that matters is unfoldability, and it is neither mixedness nor entropy in the local sense, since the checkerboard maximizes local disorder and possesses no hardness at all — it opens with three bits.

16. Work and rule reach different places

Stirring is simultaneously linear and recursive: the stroke count increments linearly while the state evolves under iteration of a fixed map. The two clocks need not remain in agreement, and they do not.

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github.com/QuHarmonics/The-Nexus-Harmonic-Reality info@quharmonics.com

strokes	local disorder	spread	versus previous point
0	0.0417	0.00000	–
10,000	0.3771	0.01861	distinguishable
50,000	0.4974	0.01796	distinguishable
200,000	0.5083	0.01329	indistinguishable
500,000	0.4898	0.01960	indistinguishable
1,000,000	0.5056	0.01263	indistinguishable
2,000,000	0.5038	0.01537	indistinguishable
4,000,000	0.4964	0.01571	indistinguishable

Table 18. Random pairwise exchange on a 24 by 24 lattice, ten seeds per point. Saturation at fifty thousand strokes; two million and four million are not separable. [MEASURED]

Three million nine hundred and fifty thousand strokes are performed after saturation, all of them real work, and the field records none of them. The stroke count is not recoverable from the state. Three ledgers must therefore be kept apart: the number of operations, the information in the state, and the physical work performed. Work can accumulate without the chosen state projection accumulating a readable record of it, and this requires no violation of any conservation law — only that many histories map to one observable state, which is the ordinary inverse-problem structure.

The second observation is sharper. Stirring saturates at local disorder 0.50. The perfect mix is at 1.00. Stirring does not approach it slowly; it does not approach it at all.

Work moves a system into the unfoldable interior. Only a rule places it on a fold. The perfect mix is a three-bit statement and no quantity of stirring reaches it.

This closes a gap that the earlier sections left open. Effort and structure are not two rates of progress toward the same destination. They terminate in different regions.

17. The cube

A permutation puzzle is the cleanest available test of whether disorder purchases distance, because the group is finite, the solution is present in every state as the inverse of the scramble, and the entire distance distribution can be computed exhaustively.

distance to solved	states	percent of space
0	1	0.000%
1	9	0.000%
2	54	0.001%
3	321	0.009%
4	1,847	0.050%
5	9,992	0.272%

distance to solved	states	percent of space
6	50,136	1.365%
7	227,536	6.193%
8	870,072	23.681%
9	1,887,748	51.379%
10	623,800	16.978%
11	2,644	0.072%

Table 19. Complete breadth-first search of the 2x2x2 group, half-turn metric. 3,674,160 states, diameter 11. The diameter reproduces the published value. [THEOREM, exhaustive]

**Three and a half million states, and no state is more than eleven moves from the solution.
More than half the entire space sits at distance exactly nine.**

There is no deep end. One cannot scramble one's way somewhere far, because far does not exist: the diameter was fixed by the group and nothing is introduced by scrambling. The difficulty is entirely the identification of which eleven moves, out of the sequences of that length, and that identification is a reader-side quantity by the accounting of Sections 8 and 9.

Disorder is also not a spectrum here. Half the space occupies a single distance value. Maximal mixing is not a destination approached by work; it is where nearly everything already is, which is the same shape as the stirring saturation of Section 16 and suggests that the saturation value of a mixing process is simply the typical distance in its group, a number that can be small.

18. What the sequence shows

Nine independent measurements have been reported, on unrelated objects, with different instruments. In every one, a cost was paid and the audit located it outside the object.

measurement	what was paid	where it landed
inverse problem, Section 7	6.15 bits then nothing	reader; floor has closed form
shelf, Section 8	$\log_2(n)$, conserved	arrangement or reader, freely
bisection, Section 9	12 history entries	reader's history, exactly $\log_2(n)$
sorting, Section 10	$n \log n$ comparisons	the comparison model
basis, Section 11	total failure to solve	reader's coordinates
ring lock, Section 12	all payoff in last ring	composability
hardening, Section 13	success rate to zero	composability, not availability
stirring, Section 16	3.95M unrecorded strokes	the stirrer
cube, Section 17	search for eleven moves	reader; diameter is eleven

Table 20. Audit of nine measurements. No field defect appears in any of them.

The recurring structure beneath all of them is the two-reading condition of Section 4. Index and count, arrangement and reader, mixedness and foldability, state and trajectory, operation count and state information: in each case one object supports two simultaneous descriptions, neither reducible to the other, and the quantity that has been mistaken for a property of the object is the difference between the descriptions. Where the two readings collapse, the object is dead — that is the wall condition. Where they are held apart, the separation is charged, and the charge is the tax.

Difficulty is a location, not a magnitude. It is the amount by which the second reading has been left unpaid, and it resides in whichever party is carrying it.

The methodological consequence is that a difficulty claim is incomplete without an audit of which party holds the ledger. A statement that a task is hard, with no column recording what the arrangement contributed, is a measurement of the reader reported as a measurement of the world.

19. Status ledger

Statement	Status
Closure prunes zero candidate laws at every n	THEOREM (enumerated, $n \leq 7$)
Wall-admissible fraction is $(n-1)^n / n^n \rightarrow 1/e$	THEOREM (brute-forced, $n \leq 7$; closed form beyond)
Wall cost equals n copies of $-\log_2(1 - 1/n) \rightarrow \log_2(e)$	MEASURED, exact to $1e-9$ over six decades
Wall condition unsatisfiable at $n = 1$; C_1 forces C_0	THEOREM
Origin-blind maps are exactly the n translations	THEOREM (brute-forced, $n \leq 7$)
Wall condition = the two index origins must not coincide	DERIVED from the above
Every admissible map on finite X is eventually periodic, no 1-cycles	THEOREM
Uniform out-degree costs positive bits, growing in n	MEASURED (exhaustive, $n \leq 5$)
Observation of a hidden law saturates; floor decomposes exactly	MEASURED (exhaustive over 15625 laws)
Off-path residual matches (unvisited) $\times \log_2(n-1)$ to all digits	MEASURED
Locate-cost conserved at $\log_2(n)$ across arrangement and reader	MEASURED
Sign gradient carries zero distance information at every scale	MEASURED
Resolving bracket width is 2, independent of n	DERIVED
Bisection changes nothing in the field; cost = $\log_2(n)$ history	MEASURED
Comparison bound is a model property; radix is linear in n	MEASURED (textbook, reproduced)
Invertible mixing collapses hill-climbing $50/50 \rightarrow 0/50$	MEASURED

Statement	Status
Key value is non-additive; half a key returns 7.5%	MEASURED
Hardening preserves ~40% informative local moves at zero success	MEASURED – corrects a prior claim by this author
Hardening is a quench (one round), not a ramp	MEASURED
Accumulator changes $k/2$ bits per quantum at every fill level	MEASURED
Checkerboard: disorder 1.0000, 160 bytes, 3-bit separation key	MEASURED
Foldability high at both disorder extremes, collapses in middle	MEASURED
Stirring saturates at 0.50 and cannot reach the fold at 1.00	MEASURED
Stroke count unreadable from state after saturation	MEASURED
2x2x2 group: 3,674,160 states, diameter 11, 51.4% at distance 9	THEOREM (exhaustive BFS)
Difficulty is a location rather than a magnitude	FRAMEWORK – relocation, not resolution
Complexity accounting lacks an arrangement column	FRAMEWORK
Hardening removes the easy directions	RETRACTED – refuted in Section 13

20. What this does not establish

The paper reports a relocation, and relocations are easy to overstate. Four boundaries are stated explicitly.

It does not resolve P versus NP

The model-locking critique of Section 10 applies to bounds derived inside a fixed access model. The P versus NP statement quantifies over all polynomial-time machines and therefore admits any reader constructible within budget. What survives is the narrower claim that the standard accounting meters the reader's work exhaustively and charges nothing for the arrangement's, and that everything living on that seam — preprocessing, indexing, amortization, data structures — is the missing column being rediscovered locally. Converting that observation into a result requires a problem class for which the arrangement column can be filled and measured, and demonstrating that the total does not move. That has not been done here.

No lock in this paper is hard

The mixing of Sections 11 and 12 is linear over $GF(2)$ and its inverse is recoverable in cubic time. The threshold curve of the ring lock is a genuine measurement of how partial alignment pays, and it is not evidence that any key is difficult to obtain. The corresponding measurement with a mixing whose inverse is not recoverable in budget has not been run, and until it is, the claim that the ring structure is the base class and nonlinearity merely a cost multiplier is a conjecture.

The reader-side result is not yet at risk

Nine measurements placed every cost outside the object, and none of them could have done otherwise, because each posed a task to a reader and audited the reader. A measurement that could falsify the pattern requires two readers with different histories disagreeing about the same collapse, so that the disagreement has somewhere to live other than in either reader's ledger. That experiment is specified and has not been performed. Until it is, the pattern should be read as a consistent accounting convention rather than as a demonstrated property of fields.

Finite computations are not promoted

Statements marked THEOREM in the ledger are exhaustive enumerations over a stated finite range together with a closed form that the enumeration confirms, or complete searches of a finite group. Statements marked MEASURED are seeded computations reported as computations. No finite computation in this paper is offered as proof of an unbounded claim, including the asymptotic identification of the wall cost with $\log_2(e)$, which is exact as a limit of the closed form and measured as an agreement of two expressions.

21. Frontier

Three computations follow directly and are small enough to run.

First, the two-reader falsification test described above, which is the only measurement in this programme's current inventory that could break the reader-side accounting rather than confirm it again.

Second, the ring-lock threshold under a mixing whose inverse is not recoverable in budget. If the threshold curve retains its shape and only the key-recovery price moves, the ring structure is the invariant and nonlinearity is a multiplier. If the curve changes shape, the two are not separable and the base-class claim fails.

Third, the conversion measurement implied by Section 7. Run the inverse problem from a second start state in a disconnected orbit, then couple the two, and check whether the drop in residual equals the off-path term exactly. If it does, the operation that couples independent trajectories has a measured exchange rate, and the identification of that operation with collision acquires a number rather than a description.

Incompleteness is stated once, here. Everything above carries its own status label, and the labels rather than the prose are what the reader should hold the claims to.